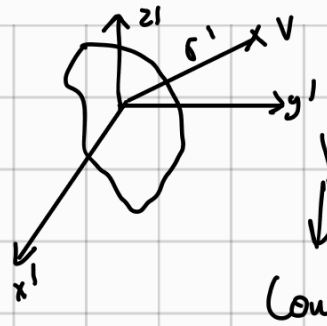


$$V = \frac{1}{4\pi\varepsilon_0}\int_{\mathcal{V}}\frac{\rho}{r'} = \frac{1}{4\pi\varepsilon_0}\iiint_{x',y',z'=\infty}^{\infty}\frac{\rho(x',y',z')}{r'}dx'dy'dz' + V_0 \rightarrow \text{sabit}$$

Coulombpotansiyeli $V(x,y,z)$

$$r' = \left[(x-x')+(y-y')+(z-z')\right]^{1/2}$$

$$\vec{E} = -\text{grad } V$$

potansiyel elektrostatik alan

$$\text{rot}\vec{E} = -\text{rotgrad } V = 0 \;//$$

$$\Delta V = \text{div grad } V \qquad \Delta = \vec{\nabla}\cdot\vec{\nabla}$$

$$= -\text{div}\vec{E} \qquad = \frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}$$

$$V = \frac{1}{4\pi\varepsilon_0}\int_{\mathcal{V}}\rho(x',y',z')\,\Delta\left(\frac{1}{r'}\right)d\mathcal{V}$$

$r' \neq 0$ ise $\Delta\left(\frac{1}{r'}\right) = 0$

ρ nun olmadığı yerde

$$\lim_{r\to\infty} V(x,y,z) \longrightarrow 0 \qquad V_0 = 0 \text{ kabul edilir}$$

Poisson denklemi

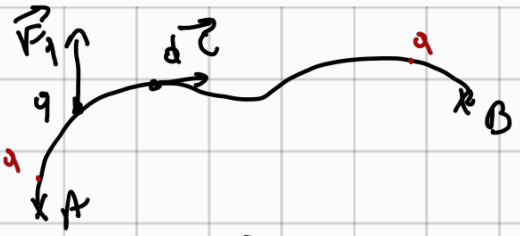
$$\Delta V = -\frac{\rho}{\varepsilon_0} \quad \text{(her yerde)}$$

$$\rho \triangleq \frac{dQ}{d\mathcal{V}}$$

$$V(x_1,y_1,z_1) = \frac{1}{4\pi\varepsilon_0}\iiint_{\mathcal{V}} \cdots + V_0$$

$$V(x_2,y_2,z_2) = // \iiint \qquad + V_0$$

$$V_1 - V_2 = \frac{1}{4\pi\varepsilon_0}\left[\iiint \cdots - \iiint\right] + V_0 - V_0 \;(=0)$$



$$\vec{F}_1 = q.\vec{E} = -q\,\text{grad}\,V = -q\,\vec{\nabla}.V$$

dışardan yapılan iş

$$\vec{F} = -\vec{F}_q = q\,\text{grad}\,V$$

$$W_{AB} = \int_A^B \vec{F}.d\vec{c} = \int_A^B q\,\text{grad}\,V\,d\vec{c}$$

$$= q\int_A^B \text{grad}V\,d\vec{c}$$

$$d\vec{c}\,\frac{\partial V}{\partial c} = \text{grad}V.\underbrace{\vec{t}.dc}_{d\vec{c}} \qquad \vec{t} = d\vec{l}$$

$$\frac{d\vec{c}}{dc} \triangleq \vec{t}$$

$$= q\int_A^B \text{grad}V.d\vec{c}$$

$$= q\int_A^B \frac{\partial V}{\partial c}.dc = qV\Big]_A^B$$

$$= qV(B) - qV(A) = q[V(B) - V(A)]$$

$q = 1\,C$ (birim yük)

$A \to \infty \quad V(A) \to 0$

$$= qV(B)$$

$$= 1\,V(B)$$

V_B: bir birim yük sonsuzdan sonlu bir B noktasına getirmeka yapılan işe o noktadaki potansiyel denir

$q = 1\,C$ yükü Adan B'ye getirilinceye kadar yapılan işe potansiyel farkı denir.

$W_{AB} > 0 \Rightarrow V_B > V_A$

↘ Enerjiyi biz harcadık anlamına gelir

Çünkü $\vec{F}$ bizim uyguladığımız kuvvet

$W_{AB} = 0$ ne zaman olur

$V_B = V(B_1)$

$V_A = V(A_1)$

$q = 0$ ' a olur

$q \neq 0$ için $V_B = V_A$ olmalıdır

$A \to \infty \quad W_{AB} = q V_B$

$$\vec{F_q} = \frac{d}{dt}[m(v)\vec{v}]$$

(küçük hızlarda

$$\vec{F_q} = m_0 \cdot \frac{d\vec{v}}{dt}$$

$$\vec{dc} \cdot \vec{F_q} = m_0 \cdot \frac{d\vec{v}}{dt} \cdot \vec{dc}$$

$$\vec{F_q} \cdot \vec{dc} = m_0 \frac{d\vec{v}}{dt} \cdot \vec{v} \cdot dt$$

$\vec{v} // d\vec{v} \Rightarrow \vec{v} \cdot d\vec{v} = v \cdot dv$

$$\int_A^B -\vec{F} \cdot \vec{dc} = \int_A^B \vec{F_q} \cdot \vec{dc}$$

$$= -q \int_A^B \text{grad} V \, \vec{dc} = -q(V_B - V_A)$$

$$= m_0 \int_A^B \frac{d\vec{v}}{dt} \cdot \vec{v} \cdot dt = m_0 \int_A^B v \, dv = \frac{1}{2} m_0 v^2 \Big|_A^B$$

$$= \frac{1}{2} m_0 (v_B^2 - v_A^2)$$

$$\boxed{q V_B + \frac{1}{2} m_0 v_B^2 = q V_A + \frac{1}{2} m_0 v_A^2}$$

Enerji korunur

* İş goldan bağımsızdır.